



City of Melbourne

Flood and Storm Sub-Plan

A Sub-plan of the Municipal Emergency Management Plan



The City of Melbourne Municipal Emergency Management Planning Committee respectfully acknowledges the Traditional Owners of the land and waters of City of Melbourne. We pay our respects to Elders past, present and emerging, and are committed to working with Aboriginal and Torres Strait Islander communities to achieve a shared vision of safer and more resilient communities.

The Flood and Storm Sub-plan has been prepared by VICSES on behalf of, and in collaboration with the City of Melbourne Municipal Emergency Management Planning Committee (MEMPC). It was prepared in accordance with, and complies with the requirements of the *Emergency Management Act 2013* including having regard to the guidelines issued under section 77, [Guidelines for Preparing State, Regional and Municipal Emergency Management Plans](#) (the **Guidelines**).

It was endorsed by the City of Melbourne Municipal Emergency Management Planning Committee on 09 December 2025 as a sub-plan to the City of Melbourne Municipal Emergency Management Plan (MEMP). A Statement of Assurance for this plan was approved by the North West Metro Emergency Management Planning Committee on 30 July 2026.

This sub-plan was published on the [VICSES website](#) on September 2026.

Any printed hard copies of this Flood and Storm Sub-plan are considered 'uncontrolled documents' and may not reflect the most current or updated version.

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1. INTRODUCTION

1.1. Aim

The aim of this MEMP Flood and Storm Sub-plan (MFSSP or the 'Sub-plan') is to detail the local arrangements for before, during and after flood and storm incidents within the municipal footprint.

1.2. Objectives

The objectives of this Sub-plan are to:

- Align with and reflect the state emergency management planning framework and wider emergency management planning arrangements and associated documentation.
- Identify and *assess the flood and storm risks* which may impact the people, property and/or the environment in the municipal area.
- Support the implementation of measures to avoid or *minimise future risks* and *reduce existing risks*, consideration to the causes and consequences of storm and flood incidents within the municipal area.
- *Manage residual risks* with a focus on how access to better information can reduce the consequences of flood events.

1.3. Authorising environment

This is an updated version of this City of Melbourne MEMP Flood and Storm Sub-plan (MFSSP). It replaces and supersedes the City of Melbourne MEMP Flood and Storm Sub-plan October 2025.

This plan is a Sub-plan of and aligns with the [City of Melbourne Municipal Emergency Management Plan \(MEMP\)](#). It is consistent with the [State Emergency Management Plan \(SEMP\)](#) and in alignment with the core capabilities of the [Victorian Preparedness Framework](#).

Comments and feedback on this Sub-plan are welcome and should be sent to: ust.nwmetro@ses.vic.gov.au.

The City of Melbourne MEMPC is the owner of this MFSSP, pursuant to *Part 6A of the Emergency Management Act 2013* (as amended).

To the extent possible, this Sub-plan does not conflict with, or duplicate, any other plans in relation to emergency management that are currently in force within Australia as per the *Emergency Management Act 2013 60AC(c)*.

Section 60AI (2) of the Emergency Management Act 2013 allows the MEMPC and Emergency Management Victoria to exclude information that is related to critical infrastructure, personal information or information that is of a commercially sensitive nature.

In accordance with its roles and responsibilities set out in the SEM, the Victoria State Emergency Service (VICSES) has prepared this plan in collaboration with the MEMPC. It considers the outcomes of the Community Emergency Risk Assessment (CERA) process undertaken by the MEMPC.

Engagement with communities through the update of this Sub-plan has primarily taken place through the Municipal Flood Plan Working Group (MFPWG), along with input from community representatives serving on the MEMPC.

Membership of the MFPWG comprises of the following representatives from the following agencies and organisations:

- VICSES Operations Officer Emergency Management
- VICSES Footscray & Port Phillip Unit Controllers
- Melbourne City Council (Municipal Emergency Management Officer)
- Victoria Police (Municipal Emergency Response Coordinator)
- Melbourne Water
- Department of Families, Fairness and Housing (DFFH)
- Fire Rescue Victoria (FRV)
- Emergency Recovery Victoria (ERV)
- Department of Health (DH)

1.4. Exercise, evaluation and review

This plan will be reviewed and updated at least every three years, with consideration given to earlier revisions as required to ensure the plan provides for a current, integrated, coordinated, and comprehensive approach to flood emergencies, and consideration of potential escalation of climate-related hazards.

Earlier reviews may be triggered by this plan being applied in a major emergency or exercise or following a flood study, a significant change in flood mitigation measures, a substantial change to relevant legislation or arrangements, including the SEMP.

1.4.1. Exercise

The MEMPC is responsible for organising exercises to test this plan, which should be conducted within the plan's lifecycle. Ideally, these exercises will be scheduled shortly before the period of highest risk for floods and storms, typically during winter and spring.

This Sub-plan was tested as part of the 2022 flood event and lessons from these events have been embedded into this plan.

Event	Exercise Type	Date
Victorian Flood event	Tested	October 2022
AFAC Maribyrnong Tour	Tour	12 March 2024
Local Boating exercises	Rescue Boat operations	Numerous dates in 2024 & 2025
Storm Events - 2023	Activation of Regional operations Centre (ROC) @ Sunshine Incident Control Centre	08/07, 11/12, 26/12
Storm Events – 2024	Activation of ROC	03/01, 01/03, 27 & 28/08
Storm Events - 2025	Activation of ROC	01/12, 23/02, 26/05, 22/10

1.5. Audience and linkages

1.5.1. Audience

The audience of this Sub-plan is local government and agencies within the emergency management sector in City of Melbourne, including business and community groups who hold a significant role in the management of the flood and storm emergencies. Although the wider community is not the primary audience, community members may find the contents of this plan informative.

1.5.2. Linkages

This Sub-plan does not operate in isolation and is integrated within the broader emergency management planning framework, which guides flood and storm management at the State, Regional, Municipal and Local level.

This plan refers to a range of existing resources relating to floods and storms and accords with the direction set through related plans and policies and provides links to where the reader can obtain further information.

Access to operationally sensitive information may require a login. This includes documents stored on the Emergency Management Common Operating Picture ([EM-COP](#)), such as [Joint Standard Operating Procedures \(JSOPs\)](#). Documents stored on the VICSES Intranet, such as Standard Operating Procedures (SOPs) and restricted information can be obtained by contacting a VICSES representative.

The sub-plan is consistent with and supports:

- [State Emergency Management Plan](#) (SEMP)
- SEMP [Flood Sub-plan](#), SEMP [Storm Sub-plan](#)
- [North West Metro \(Central\) Regional Emergency Storm Plan](#)
- [North West Metro \(Central\) Regional Emergency Flood Plan](#)
- [The Municipal Emergency Management Plan \(MEMP\)](#)
- [Community Local Flood Guides \(LFGs\)](#):
 - ❖ City of Melbourne LFG
 - ❖ Kennington, North Melbourne and West Melbourne

1.5.3 Restricted Operational Information

The following documents support and supplement this plan. The Annex's listed below contain restricted operational or sensitive information and can be obtained through a VICSES representative as required:

Restricted document(s)
Flood Intelligence and VICSES Response Arrangements for Flood and Storm in the City of Melbourne:
■ Annex A – Flood and Storm Threats for Municipality
■ Annex A1 – Flood evacuation arrangements (if applicable)
■ Annex B – Typical flood peak travel times
■ Annex C – Localities and At-Risk Communities
■ Annex D – Maps and Schematics
■ Annex E – Storm Damage and Specific Response Arrangements
■ Annex F – Flood and Storm Warning Systems

These Annexures may contain operational or sensitive information and may be obtained through a VICSES representative as required.

1.5.4 Hyperlinks

Documents or resources that are referred to frequently throughout this plan (such as the [SEMP](#)) may not be hyperlinked in each instance.

All hyperlinks were accurate at time of publication and currency of the linked content remains the responsibility of the host agency.

2. THE EMERGENCY CONTEXT

2.1. Risks

The effects that intense storms can have on the City of Melbourne can be far reaching due to the amounts of road, rail and freight infrastructure and the dense concentration of people, including residents, workers and tourists. The ability for residents and workers to move in and out of the CBD can be greatly impacted. As many large events and festivals are also held within the Municipality, there is the potential for large amounts of people to be impacted at one time.

The built-up nature of the City of Melbourne makes it susceptible in Severe Weather Events. Older drainage infrastructure unable to cope with flash flooding volumes, combined with roads and buildings interrupting overland flow paths will divert floodwaters, putting property at risk. Mature trees located within the municipality may also create traffic hazards or building damage in storm events.

This Appendix details areas of risk from severe weather events by requests for assistance to the Victoria State Emergency Service (VICSES).

2.1.1 Riverine flooding

Large severe riverine floods typically occur when moist, warm airflow from northern Australia brings moderate to heavy rainfall over a 12-hour period or more. These events are usually preceded by extended periods of rainfall that saturate the catchment, fill on-stream dams, and reduce the capacity of natural floodplain storage, resulting in increased runoff during the subsequent period of heavy rainfall.

Less severe but still significant flooding can also result from a series of cold fronts during winter and spring. These systems gradually wet the catchment and increase water levels in dams and waterways, with major flooding occurring when prolonged moderate to heavy rainfall follows.

2.1.2 Flash flooding

Short Duration, high intensity rainfall (usually associated with thunderstorms) can also cause localised flooding within the municipality, along overland flow paths when the local urban drainage system surcharges. Such events, which are mainly confined to the summer months, do not generally create widespread flooding since they only last for a short time and affect limited areas. Flooding from these storms occurs with little warning and localised damage can be severe.

High intensity rainfall such as associated with thunderstorms giving average rainfall rates of more than 20mm/hour for an hour or more is likely to lead to flash flooding and / or overland flows, across the urbanised parts of the municipality.

Blocked or capacity impaired stormwater drains can also lead to overland flows and associated flooding: the drain surcharges and excess water flows above ground.

Flooding within City of Melbourne catchments is generally caused by short duration thunderstorm events because these produce the highest rates of runoff in hard-lined drainage systems serving relatively small and highly impervious catchments. Planning Scheme delineations within the Municipality for Flash Flooding events include Special Building Overlays 1 & 2 (SBO1 & SBO2). Contact Melbourne City Council for information on these planning scheme overlays.

2.1.3 Severe weather and storms

Severe thunderstorms and associated extreme weather conditions, such as tornadoes or microbursts, can have significant impacts on the community, built environment and essential services. While severe thunderstorms may affect large areas, tornadoes and microbursts in Victoria are typically highly localised, often impacting only one or two suburbs. Despite their limited geographic footprint, these events can cause intense and concentrated damage.

Typical consequences of severe storm events include:

- Wind damage to residential and commercial buildings
- Uprooted or fallen trees causing property damage and blocking roadways
- Localised flash flooding
- Large hail
- Road damage and temporary road closures
- Power outages
- Disruption to telecommunications services
- Discharge/overflow of sewer systems
- Impacts on critical infrastructure (e.g. health, water, energy, transport)
- Entrapment of people in homes or vehicles due to fallen trees or floodwater
- Epidemic Thunderstorm Asthma (ETSA)

2.1.4 Tidal flooding and storm surges

Moderate to heavy rainfall, coupled with a high or incoming tide from Port Phillip Bay can exacerbate flooding within the municipality or create areas of flooding in and around the drainage network. Due to the proximity of the Municipality to Port Phillip Bay and its flat terrain, tidal flows from Port Phillip Bay may reduce the capacity of the stormwater drains to discharge runoff back into the bay, while extreme storm events can cause backflow to the point where water surcharges back above ground around the drainage pits and channels.

2.2. Municipality Major Waterways, Drains and Dams

The City of Melbourne municipality is located at the bottom of the Port Phillip Bay catchment where waterways are saline, groundwater is shallow, and pollution occurs from upstream sources¹. The City of Melbourne municipality contains three major waterways: the Yarra River; the Maribyrnong River; and Moonee Ponds Creek. See Mapping in **Annex D (restricted)**.

The Yarra River begins its journey in the Yarra Ranges to the east of Melbourne and runs through the Yarra Valley and Melbourne's eastern suburbs before dissecting the City of Melbourne municipality and discharging into Port Phillip Bay. The Yarra River drains much of eastern and northern Melbourne with the City of Melbourne municipality constituting less than one per cent of its entire catchment area. See Yarra River Catchment Schematic in **Annex D (restricted)**.

¹ Melbourne Water (2018): City of Melbourne Flood Management Plan

The Maribyrnong River catchment covers 1,430 square kilometres; it rises to the north-west of Melbourne and drains into the Yarra River. The Maribyrnong River is the western border of the City of Melbourne municipality. See Maribyrnong River Catchment Schematic in **Annex D (restricted)**.

The Moonee Ponds Creek catchment covers a total area of 145 square kilometres and flows through Melbourne's northern suburbs before entering the City of Melbourne municipality². The Moonee Ponds Creek flows into the Yarra River upstream of Appleton Dock through a realigned channel beside the Bolte Bridge. See Moonee Ponds Creek Catchment Schematic in **Annex D (restricted)**.

Other waterways or drainage systems within the City of Melbourne municipality include Royal Park Wetland, the Dynon Road Tidal Canal, and Hanna Street Main Drain².

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Melbourne Water Drains & Waterways	Suburb/s	Melbourne Water Drains & Waterways	Suburb/s
Arden Street Main Drain	Parkville & North Melbourne	Johnson Street Main Drain	Docklands
Cowderoy Street Main Drain	Melbourne	Maribyrnong River	Flemington, Kensington & West Melbourne
Dynon Road Tidal Canal	West Melbourne	Moonee Ponds Creek	Parkville, North Melbourne, Kensington, West Melbourne & Docklands
Elizabeth Street Main Drain	Melbourne	Royal Park Main Drain	Parkville
Gittus Street Main Drain	Docklands	Yarra Park Main Drain	Melbourne
Hanna Street Main Drain	Melbourne & Southbank	Yarra River	Melbourne, Southbank, South Wharf, Docklands, West Melbourne & Port Melbourne

Table A1 – Melbourne Water Drains and Waterways within the City of Melbourne

2.2.1 Dam Spilling / Failure

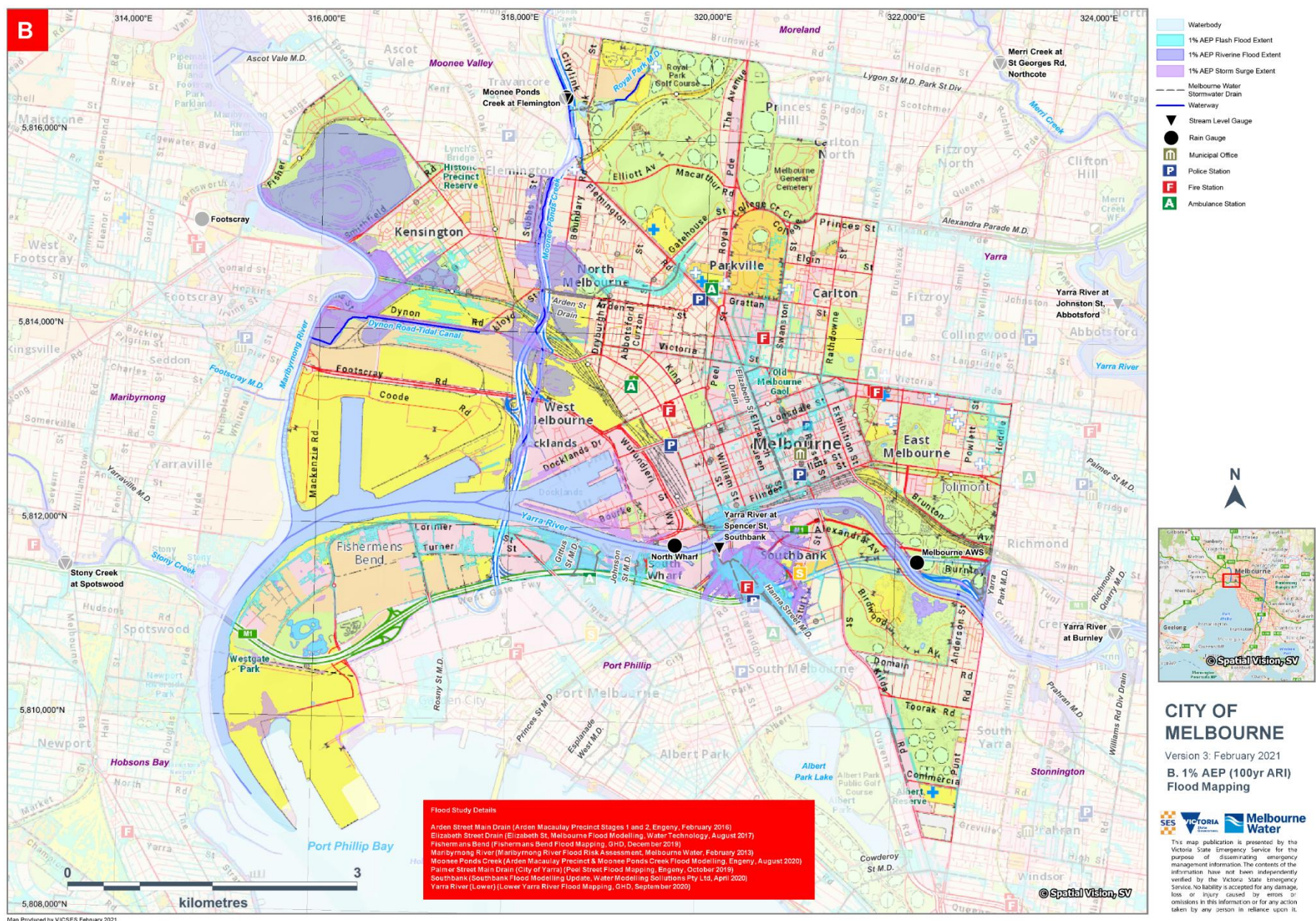
Flooding resulting from failure of the following dams is likely to cause significant structural and community damage within the City of Melbourne. See Dam Failure in Section 3 of this plan for more information. Note that if the storage capacity is reached and water flows over the spillway, this is not to be referred to as a flow release or a storage breach or failure.

Dam	Location	Owner	Dam Capacity	Full Supply Level	Melway Reference
Greenvale Reservoir	Greenvale	Melbourne Water	27,195 ML	167.12m AHD	179 D6
Rosslynne Dam	3km northwest of Gisborne	Southern Rural Water	25,400ML	450.9m AHD	VicMap Book 6443 F1
Sugarloaf Reservoir	Christmas Hills	Melbourne Water	93,411 ML	178.00m AHD	273 E6
Upper Yarra Reservoir	Reefton	Melbourne Water	200,051 ML	366.53m AHD	X912 U2

Table A12 – Large Dams around Greater Melbourne with flooding consequence risks within the City of Melbourne

DEECA has developed the [Dam Safety Response Plan](#) to assist in managing its obligations associated with dam safety incidents.

For further information in relation to the above risks within the municipality refer to **Annex A** in the Flood & Storm Intelligence and VICSES Response Arrangements for the City of Melbourne (restricted).



3. MANAGING FLOOD AND STORM EMERGENCIES

3.1 State Emergency Management Priorities

The State Emergency Management Priorities that are outlined in the [SEMP](#) guide all decisions before, during, and after any emergency, and apply to all aspects of this plan.

3.2 Roles and Responsibilities

An agency that has a role or responsibility under this plan must act in accordance with the plan.

There are a few agencies with specific roles that will act in support of VICSES and provide support to the community in the event of a serious storm or flood within the City of Melbourne. These agencies will be engaged through the Incident Emergency Management Team (IEMT).

The general roles and responsibilities of supporting agencies are as detailed in the [MEMP](#), [SEMP](#), [SEMP Flood Sub-plans](#) and [SEMP Storm Sub-plan](#).

3.3 Mitigation and Preparedness

3.3.1 Floodplain Management Strategy

The Victorian Floodplain Management Strategy sets the policy and accountability framework for flood related mitigation activities in Victoria. The [Floodplain Management Strategy for Port Phillip and Westernport](#) sets out the implementation of flood mitigation activities for the Region, including for the City of Melbourne.

3.3.2 Structural flood mitigation measures

Structural flood mitigation measures are any physical construction to reduce or avoid possible impacts of flood hazards, or the application of engineering techniques or technology to achieve flood hazard resistance and resilience in structures or systems.

For more detailed Flood Intelligence and VICSES Response arrangements surrounding structural flood mitigation measures refer to **Annex C** in the Flood Intelligence and VICSES Response Arrangements for Flood and Storm in City of Melbourne (restricted).

3.3.3 Planning Scheme Controls

Flood mitigation within the municipality is supported through planning scheme controls such as the Land Subject to Inundation Overlay (LSIO), Floodway Overlay (FO), and Special Building Overlay (SBO), which guide development in flood-prone areas. These overlays help manage flood risk by ensuring appropriate building siting, floor levels, and drainage requirements. The planning controls work alongside building

regulations and the National Construction Code to support safer, more resilient development across the municipality.

Planning overlays play a critical role in reducing flood risk by mandating floor levels above the predicted flood level, requiring new developments to incorporate adequate drainage measures, and protecting overland flow paths from being obstructed.

Council is proactively applying these planning tools to assess new proposals, regularly updating overlays as new flood modelling becomes available, and advocating for stronger planning provisions across Victoria to further support community resilience.

In addition to guiding new development, planning controls also respond to changes in flood risk identified through updated and improved flood modelling. This modelling can reveal new risks to existing properties. Climate change is expected to further increase flood risk across the municipality, expanding areas subject to inundation and increasing depths for existing at-risk properties.

3.4. Community Resilience

A resilient community is socially connected and has infrastructure that can withstand disaster or shocks and foster community recovery. Resilient communities promote individual and community wellbeing, use of available resources and cohesiveness to strengthen their communities for everyday, as well as extreme challenges.

3.4.1 Flood and storm specific community education and engagement

Community education and engagement is a vital component in the community understanding risks and preparing for emergencies. The development of relevant and appropriate community resources and activities empower the community and enhance their resilience through being well informed and therefore being equipped (emotionally and physically) for an emergency.

Flood and storm risks cannot be fully mitigated, and a shared responsibility is vital. VICSES, with the support of the City of Melbourne Council and other agencies, will coordinate and deliver targeted community flood and storm engagement programs by:

- Providing accurate preparedness information through media, publications and engagement activities
- Facilitating and supporting community programs, events and targeted initiatives
- Offering technical advice to ensure resources remain accurate and relevant
- Collaborating with schools, businesses, and organisations to strengthen planning and preparedness
- Contributing to exercises, forums, and community engagement opportunities
- Acting as liaisons during events to ensure clear and consistent communication

Community members are encouraged to:

- Understand their flood and storm risk
- Prepare by developing a home emergency plan and emergency kit (see the [VICSES website](#))
- Stay informed and follow emergency services' advice during warnings
- Engage in community programs, events and resources such as Local Flood Guides, targeted initiatives, and Community Emergency Management Plans (CEMPs).
- Participate in emergency planning at facilities such as schools, retirement villages and businesses

3.4.2 Public information and warnings for flood and storm emergencies

The SEMP Flood Sub-Plan and the SEMP Storm Sub-Plan detail the arrangements for public information and warnings for flood and storm. VICSES [SOP008 Severe Weather Notification and Activation Process](#) and VICSES [SOP009 Flood Notification and Activation Process](#) outline in detail the actions that VICSES will undertake upon receipt of a Severe Weather Warning or Flood Watch/Flood Warning from the Bureau of Meteorology.

Details on warnings issued by VICSES through [VicEmergency](#) and VICSES channels, including for areas or impacts not covered by a Bureau warning, are outlined in **Annex F** in the Flood & Storm Intelligence and VICSES Response Arrangements for the City of Melbourne (restricted).

The Incident Controller (IC), through the Public Information Unit established at the Incident Control Centre (ICC), will manage public information and warnings. If the ICC is not established, the VICSES Regional Duty Officer (RDO) and VICSES Regional Agency Commander (RAC) are responsible for managing all public communication. The City of Melbourne will work with the IC/VICSES to assist with the dissemination of public messaging and/or warnings to ensure that consistent and timely messaging occurs.

The table below summarises the flood and storm warnings issued on VicEmergency. Visit the [Australian Warning System \(AWS\) website](#) for further information.

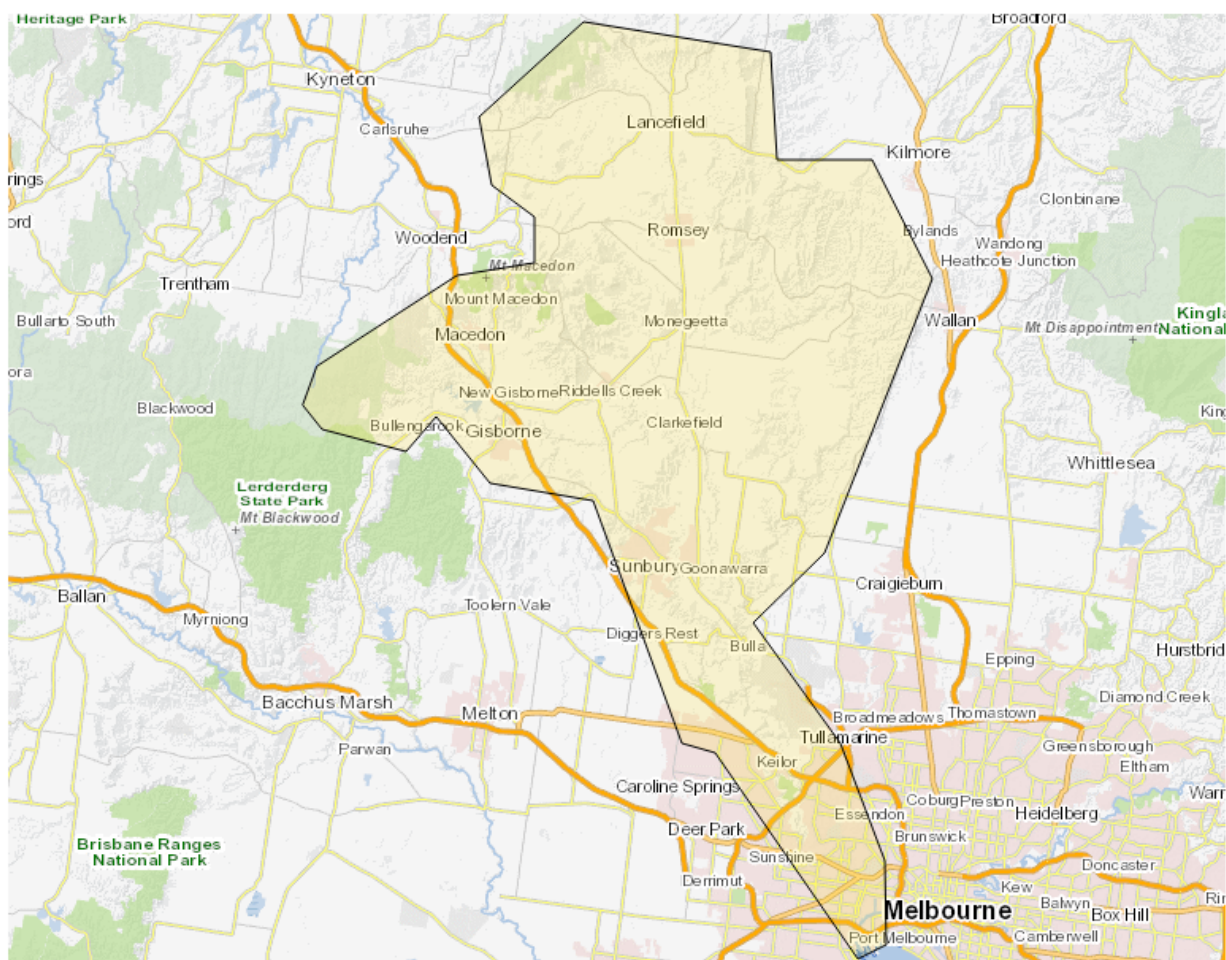
Flood Symbol	Storm Symbol	Warning Level	Meaning	Agency/Community Action
		Advice (yellow)	Flood or storm possible; no immediate danger.	An incident has started. There is no immediate danger. Stay up to date in case the situation changes.
		Watch and Act (orange)	Flooding or severe storm likely; may impact safety and property.	There is a heightened level of threat. Conditions are changing and action is required now to protect community safety.
		Emergency Warning (red)	Severe flooding or dangerous storm occurring or imminent.	An emergency warning is the highest level of warning. Action needs to be taken immediately, any delays puts life at risk.

3.4.3 BoM Flood Warning Example

ADVICE - FLOOD

Incident Location: Maribyrnong
Incident Name: MaribyrnongFloodSept2016
Issued: Set at publish time
Next Update Expected:

Map



Message

This **Minor Flood Warning** is being issued for Maribyrnong River.

- The Maribyrnong River catchment has received rainfall averaging about 31mm since 0900am yesterday. Rainfall totals of 5mm have been forecast for the catchment in the next 2 hours.
- Water levels of the Maribyrnong River and its tributaries at various locations are rising in response to the rain.
- The level of the Deep Creek at Darraweit Guim is currently 5.41m and rising. It is expected to peak above the Minor Flood Level (5.50m) this morning.
- Minor flooding in the Deep Creek and Maribyrnong River catchment is expected to affect low lying areas adjacent to the waterway. Minor roads may be closed.

The river heights at 08.14am 14/09/2016 were:

- Deep Creek at Doggetts Bridge, Lancefield: 2.22 metres, rising
- Deep Creek at Darraweit Guim: 5.47 metres, falling
- Deep Creek at Konagaderra: 3.62 metres, falling
- Bolinda Creek at Clarkefield: 1.19 metres, rising
- Deep Creek at Bulla: 2.39 metres, falling
- Rosslynne Reservoir, Head Gauge: 38.52 metres, rising
- Jacksons Creek at Sunbury: 2.13 metres, rising
- Steele Creek at Keilor East: 1.19 metres, rising
- Maribyrnong River at Keilor North: 3.58 metres, rising
- Maribyrnong River at Keilor: 1.84 metres, rising
- Maribyrnong River at Maribyrnong: 0.04 metres, rising

Stay informed - monitor your local conditions and remain alert.

What you should do:

- Be prepared to act if your situation changes.
- You should stay informed by listening to emergency broadcasters and monitoring warnings.
- Monitor weather forecasts and river levels. Go to www.bom.gov.au/vic/warnings.
- Floodwater is dangerous - never drive, walk or ride through floodwater.

Impacts in your area:

- Flooding above floor level of a single story home is likely to occur in some locations.
- Local roads may be closed and low bridges may be underwater.
- Areas around rivers and streams may be flooded.

This message was issued by State Emergency Service.

The next update is expected by 4PM this afternoon or as the situation changes.

Flood information:

- For river heights check www.bom.gov.au or phone 1300 659 217.
- For urgent animal welfare issues call [Agriculture Victoria](http://www.agriculture.vic.gov.au) on 136 186 or your local vet.

3.4.4 Flood Intelligence

Flood intelligence supports decision making and planning for flooding by providing reliable and accurate information relating to:

- The expected level, depth, and velocity of floodwater and its consequences.
- Determination of actions to be undertaken in response to the identified consequences.

The Department Energy, Environment and Climate Action (DEECA) maintains the [FloodZoom](#) flood intelligence platform. Inquiries regarding FloodZoom access should be directed to accounts@floodzoom.vic.gov.au.

For more detailed Flood Intelligence and VICSES Response arrangements refer to **Annex C** in the Flood Intelligence and VICSES Response Arrangements for Flood and Storm in City of Melbourne (restricted).

3.4.5 Cross-border Collaboration and Area of Operations

The North West Metro Emergency Management Collaboration may recognise there is a high proportion of similar or shared flood and storm issues, risks, and memberships leading to opportunities for cross-border municipal collaboration.

The area of operations concept is detailed within the [SEMP](#) and provides flexibility to managing flood and storm response when it is geographically located across several municipalities where it encourages collaboration between municipalities as part of the Victoria's [emergency management planning framework](#) and the [Victorian Preparedness Framework](#).

3.5. Local Knowledge

As the control agency for flood and storm in Victoria, VICSES is committed to ensuring the incorporation of local knowledge in decision making before, during and after incidents. This is guided by the [VICSES policy 10.02 Local Knowledge](#).

Local knowledge is a critical element of planning. The community and other organisations can provide valuable local information about hazards, incidents and how they may evolve. This information is commonly referred to as local knowledge. This plan aims to ensure that planners and responders capture appropriate local knowledge before, during and after incidents.

As an incident escalates from local control to a larger incident management structure, it is essential that local knowledge capability is retained within the overall structure. This should include how local subject matter experts are embedded into divisional and sector command structures.

There are no formal Community Flood Observer Programs within the municipality, however local knowledge has been incorporated into this plan through consultation with local response agencies.

Local Knowledge should also include an understanding of cross-boundary impacts and shared resource demands between neighbouring municipalities, particularly where flood risk complexities may influence operational response and recovery activities.

3.5.1 Field Observers

Field Observers provide local knowledge to VICSES and the Incident Control Centre regarding local insights and the potential impacts and consequences of an incident and may assist with the dissemination of information to community members.

VICSES have trained Field Observers who can be deployed to gather intelligence in areas impacted by both flood and storm events. To support intelligence gathering during flood and storm operations, drone support can be provided by other agencies, with requests facilitated through the VICSES SAC and SCC Air Desk.

[VICSES SOP073](#) outlines the role of the Field Observer and process for their activation which is via the VICSES RDO.

3.6. Response (including relief)

3.6.1 Activation of Response

VICSES may be notified of flood and storm incidents through several sources, but the most common source is calls received via 132 500 or if the emergency is life threatening, Triple Zero (000). Other sources are via other emergency management agencies and local government. In most cases, these events are of a small scale (a level 1 incident), which local VICSES units manage without significant outside support.

In the case of more significant level 2 (regional level) or level 3 (an incident that has high complexity and may have statewide implications) Flood and storm response arrangements may be activated by the RDO VICSES North West Metro Region or RAC.

The VICSES IC/RDO/RAC will activate agencies as required as documented in the SEMP [Flood Sub-plan](#) or SEMP [Storm Sub-plan](#).

3.7. Concept of Operations

The SEMP defines the responsibilities of the Incident Controller (IC), Incident Control Centre (ICC), Incident Management Team (IMT) and Incident Emergency Management Team (IEMT).

3.7.1 Control

Sections 5(1)(b) and 5(1)(c) of the [Victoria State Emergency Service Act 2005](#) detail the authority for VICSES to respond to storms and floods.

The Role Statement within the SEMP identifies VICSES in its response functions as the [Control Agency for flood and storm](#). It identifies DEECA as the [Control Agency responsible for dam safety as well as reticulated water and wastewater \(sewerage\) service](#).

All flood and storm response activities within the City of Melbourne including those arising from a dam failure or retarding basin / levee bank failure incident will therefore be under the control of the appointed Incident Controller, or delegated representative.

3.7.2 Incident Control Centre (ICC)

The ICC may be activated in advance based on the severity of warnings and in accordance with VICSES readiness arrangements:

- [VICSES readiness and activation levels – flood.](#)
- [VICSES readiness and activation levels – severe weather.](#)

3.7.3 Escalation

Resourcing and event escalation arrangements are described in the [SEMP](#).

3.8. Considerations for operating with other agencies and stakeholders

As other agencies are deployed to assist the IC should consider the following actions.

- Establish a communications plan to enable the tasking of other agency resources which should be included in briefings.
- Embedding and determining an agreed response to downed powerlines as this is often a predominant hazard for storm events.
- Ensuring other agency personnel who are undertaking the Emergency Management Liaison Officer (EMLO) roles have access to EM-COP.
- Preparing a briefing to support in-coming other agency resources to identify:
staging area location and any safety issues with accessing it (closed roads/powerlines down)
resources available such as re-supply of consumables (tarps/sandbags)
welfare arrangements and
duty time limitations (these should be consistent with VICSES SOP 003 Fatigue Management and Duty Time Limitations)

3.8.1 Natural and cultural considerations for response

Response activities will identify and consider areas of cultural significance. Where emergency works have the potential to impact culturally sensitive sites, the Incident Controller will seek advice from a Country Advisor. Given that these events often affect areas of high cultural sensitivity, early engagement is important to inform decision-making. The most significant impacts are typically experienced during the recovery phase (refer to the recovery section below).

3.9. Initial Impact Assessment

As part of the [Victorian Preparedness Framework](#), the [SEMP](#), [SEMP Storm sub-plan](#) and [SEMP flood sub-plan](#) outline agency responsibilities and core capabilities for flood and storm impact assessments.

3.10 Storm Response Arrangements

In the initial response phase, units will receive Request for Assistance (RFA's) direct from Triple Zero Victoria and will typically respond in a business-as-usual mode, typically attending events in order of receipt or priority. This is in accordance with the [VICSES Operations Management Manual](#).

For more detailed storm damage specific response arrangements refer to **Annex E** in the Flood & Storm Intelligence and VICSES Response Arrangements for the City of Melbourne (restricted) and the City of Melbourne MEMP.

3.11 Evacuation Arrangements

Under the SEMP, Victoria Police (VicPol) has the responsibility for evacuation (Evacuation Manager) – in consultation with the control agency and other expert advice.

The IC is responsible for making a decision in consultation with VicPol as to whether evacuation is a safe option for communities and individuals.

In making this decision the IC may seek advice from other agencies or communities, as detailed in [JSOP 03.12 – Planned Evacuation for Major Emergencies](#).

3.12 Response to flash flooding

Flash flooding can be defined as flooding that occurs within six hours or less of the flood-producing rainfall within the affected catchment. This may result in isolation of individuals and communities as time to warn and respond to flash flooding is limited. The safest place to be in a flash flood is well away from the affected area. Accordingly, pre-event planning for flash floods should commence with an assumption that evacuation is the most effective strategy, provided evacuation can be safely implemented.

Emergency management response to flash flooding should be consistent with the [SEMP Storm Sub-plan](#).

3.13 Flood Rescue

Under Victoria's emergency management arrangements, rescue is considered separately to the relocation of people who are stranded or isolated by floodwater. Where the water is either fast or swift flowing and/or the people being assisted are facing actual or threatened danger of physical harm, the response escalates from relocation to rescue.

Under the [SEMP Response table 9](#) the control agency for rescue from land and water is VicPol, who operate the State Police Search and Rescue Coordination Centre (SPSARCC). VICSES is a support agency for search and rescue on land and water evacuations and incidents involving mass casualties.

To conduct water rescues during a flood, VicPol will often require assistance of appropriately trained and equipped personnel from support agencies including VICSES, Volunteer Marine Search and Rescue

Organisations, CFA, FRV and Life Saving Victoria. VicPol will coordinate with these agencies to ensure operational readiness for activation.

On occasion, VicPol may opt to respond a field capability of its rescue coordination centre to a location near the emergency. It may also work with the Triple Zero Victoria to deploy its dispatch capability to the same location to enhance rescue coordination and dispatch. Details in this plan may assist VicPol and Triple Zero Victoria in undertaking this function in the field or from the primary SPSARCC.

Refer to **Annex C** for known high-risk areas/communities (such as low-lying islands) where rescues might be required in the Flood & Storm Intelligence and VICSES Response Arrangements for the City of Melbourne (restricted).

3.14 Aircraft Management

Aircraft can be used for a variety of purposes during flood operations including evacuation, resupply, reconnaissance, intelligence gathering and emergency travel.

3.15 Animal welfare

Matters relating to the welfare of livestock and companion animals are to be referred to DEECA - [Agriculture Victoria](#).

Emergency fodder may be available. Requests for emergency fodder for livestock can be referred to DEECA - Agriculture Victoria.

Matters relating to the welfare of wildlife can be referred to [DEECA - Wildlife](#).

The [Victorian Emergency Animal Welfare Plan](#) further outlines the roles and responsibilities for emergency animal welfare in Victoria.

3.16 Essential community infrastructure and property protection

Essential community infrastructure and property such as residences, businesses, roads and utilities may be affected in the event of a flood.

VICSES maintains a stockpile of sandbags, Melbourne City Council does not supply them. The IC will determine the priorities related to the use of sandbags, which will be consistent with the strategic priorities.

The [VICSES Operations Management Manual](#) sets out the principles for sandbag use and allocation to the community. These principles do not apply to the use of sandbags by VICSES to construct and/or alter a levee. Refer to [Sandbag filling and collection point guide](#) and [SOP036 Construction, Removal or Altering of Levee and Removal of Debris](#) for further detail.

The IC will ensure that owners of essential community infrastructure are kept advised of the flood situation. Essential community infrastructure providers must keep the IC informed of their status and ongoing ability to provide services.

The VICSES website has the most current [sandbag guidelines](#) or download it from the IMT Toolbox in [EMCOP](#)-Operations.

For more specific details of essential infrastructure requiring protection and location of sandbag collection point(s) refer to **Annex C** in the Flood & Storm Intelligence and VICSES Response Arrangements for the City of Melbourne (restricted).

3.16.1 Disruption to services

Disruption to services other than essential community infrastructure and property can occur in flood and storm events.

For more detailed arrangements of likely disruption to services and proposed arrangements to respond to service disruptions refer to **Annex C** in the Flood & Storm Intelligence and VICSES Response Arrangements for the City of Melbourne (restricted).

3.16.2 Road Notifications

City of Melbourne and the Department of Transport and Planning (DTP) will continue to carry out their formal road management functions where safe to do so, including surveillance, placement of warning signs, and supporting overall traffic management arrangements in conjunction with VICPOL and the Incident Controller.

City of Melbourne staff should also liaise with and advise DTP as to the need or advisability of erecting warning signs and/or of closing roads and bridges under its jurisdiction. DTP are responsible for designated main roads and highways and councils are responsible for the designated local and regional road network.

DTP and City of Melbourne will provide community information direct to the public regarding road closures. Information will be updated on the [VIC Traffic](#) website.

For more detailed arrangements of potential road closures refer to **Annex C** in the Flood and Storm Intelligence and VICSES Response Arrangements for Flood and Storm in City of Melbourne (restricted).

3.16.3 Traffic Management Plan

Traffic management will be undertaken in accordance with JSOP 03.10 Traffic Management at Major Emergencies. The IC, who has overall responsibility for incident control, will consult with Victoria Police and, where possible, the responsible road authority to develop, implement, and monitor a Traffic Management Plan. Within City of Melbourne, DTP is responsible for main roads and highways, while City of Melbourne Council is responsible for local and regional roads, bridges, and trails. Victoria Police will coordinate the restriction of access to affected areas as directed by the IC and may support the enforcement of closures and roadblocks.

Traffic management arrangements must ensure safe egress for affected residents and businesses, and access for emergency services. For emergency services, this will include effective movement of vehicles along the length of the flooded area.

3.17 Access to technical specialists

VICSES manages contracts with private specialists who can provide technical assistance in the event of flood operations or geotechnical expertise. The Regional Controller or delegate (e.g.: VICSES RDO or IC) should make a formal request to the VICSES State Duty Officer (SDO) for the engagement of a Flood Analyst or Technical Specialist. [VICSES SOP061](#) outlines this detail, including the procedure for accessing technical specialists.

- These services may be required particularly where Catchment Management Authority or Local Government staff are unable to assist in an ICC. Specifically, they can provide the following key professional services:
- Flood intelligence (Flood Analysts).
- Geotechnical assessments..

3.18 Relief

Relief is the provision of assistance to meet the essential needs of individuals, families and communities during and in the immediate aftermath of an emergency.

Suitable relief facilities identified for use during storms and/or floods, along with the associated relief arrangements are considered restricted information. This information can be obtained by agencies through the City of Melbourne Council Senior Emergency Management Coordinator (SEMC) or the Municipal Emergency Response Coordinator (MERC). Community will be advised of the location and relevant details once a facility has been activated during a flood and/or storm event.

As per the [role statement for municipal councils](#) within the SEMP, municipal councils are responsible for municipal relief coordination.

3.18.1 Activation of emergency relief

Refer to the [SEMP Roles and Responsibilities - Relief](#) for more detail of services that may be provided and the responsible coordinating agencies.

Arrangements for relief from a flood and/or storm event within the City of Melbourne is detailed in the City of Melbourne [MEMP](#)

3.18.2 Resupply

Communities, neighbourhoods or households can become isolated during floods and in some cases, storms. This can be because of road closures or damage to roads, bridges and causeways. Under such circumstances, the need may arise to resupply isolated communities/properties with essential items.

When predictions/intelligence indicates that communities, neighbourhoods and/or households may become isolated, VICSES will advise businesses and/or households that they should stock up on essential items.

After the impact, VICSES can support isolated communities through assisting with the transport of essential items to isolated communities and assisting with logistics functions.

Resupply operations are to be included as part of the emergency relief arrangements with VICSES working with the relief agencies to service communities that are isolated.

For more detailed arrangements/information refer to **Annex C** in the Flood Intelligence and VICSES Response Arrangements for the City of Melbourne (restricted).

3.19 Recovery

As per the [role statement for municipal councils](#) within the SEMP, municipal councils are responsible for coordinating local level recovery activities. They are also the lead agency to coordinate post emergency needs assessment to determine long term recovery needs (Post Emergency Needs Assessment).

Arrangements for recovery from a flood and/or storm event within the City of Melbourne is detailed in the City of Melbourne [MEMP](#).

3.19.1 Transition from response to recovery

The [SEMP](#) sets out the transition to recovery arrangements. During the response phase, the IC will ensure they develop a plan for transition from response to recovery. The IC at the municipal tier should take a lead role in facilitating transition to recovery, working with the Municipal Recovery Manager (MRM), as it marks the end of the response phase which the Controller leads and manages.

3.19.2 Environmental and cultural heritage recovery

Local recovery planning will recognise and respect culturally significant places, Country, and the health and wellbeing of Aboriginal communities. Recovery activities should support the involvement of Traditional Owner organisations and Cultural Heritage Advisors to enable culturally informed approaches that minimise harm to Country, culture, and community.

At the municipal level, Council coordinates recovery activities and will engage with the Traditional Owner organisations as appropriate. This supports culturally informed recovery outcomes and aligns with Victoria's commitment to self-determination.

Emergency Management Victoria (Relief and Recovery) provides regional coordination and may support engagement with Traditional Owners, particularly during larger or multi-municipality events. DEECA is the lead agency for environmental recovery and works with relevant stakeholders, including Traditional Owners, to support environmental restoration and rehabilitation.

The integration of Traditional Owner knowledge, cultural values, and practices within flood recovery planning will continue to evolve as partnerships, policy, and guidance develop over time

3.20 After action review – Lessons Management

Lessons management is the critical process of learning from how we worked before and during an event, to improve the system for next time.

Depending on the size and scale of the flood or storm event, VICSES will normally coordinate a debrief or after action review of flood or storm operations as soon as practical following an event. Under the [VicPol SEMP role statement](#), it is the responsibility of the Municipal Emergency Response Coordinator (MERC) to ensure that this occurs.

When the flood is being managed as a level 3 event, Emergency Management Victoria (EMV) in consultation with VICSES may assume responsibility for debriefing.

All agencies involved in the flood incident should be represented at the debrief or after action review.